Benchmarking and Adapting Open-Source Vision-Language Models for TextVQA: A Funnel Evaluation and LoRA Fine-Tuning Study

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Abstract

TextVQA requires a model to read scene text, ground it in visual context, and answer short natural-language questions under exact-match evaluation. We present a staged empirical study that benchmarks open-source vision-language models (VLMs) under a funnel protocol designed to preserve validation integrity while still supporting broad exploration. The protocol combines stratified internal-dev screening, finalist reruns on the official validation split, winner-conditioned LoRA fine-tuning, and post-training adapter evaluation under a reproducible local-plus-remote execution boundary. Across four open-source VLM families and OCR-aware prompting variants, Qwen2.5-VL-3B emerges as the clear zero-shot winner, reaching 81.05% internal-dev accuracy and 78.72% on validation with a short-answer prompt. A subsequent 12-run LoRA study raises the final validation score to 84.76%, a gain of 6.04 accuracy points over the strongest zero-shot finalist, with parallel gains in consensus accuracy, F1, and ROUGE-L. The trained-model analysis further shows that adaptation gains are concentrated in OCR-grounded cases rather than OCR-free questions, while the winner-conditioned follow-up runs reveal only marginal sensitivity to OCR toggling after adaptation and a modest but consistent gain from scaling the training pool. Beyond the headline result, the paper contributes a defendable evaluation funnel, a compute-aware multi-stage study design, and a complete reproducible artifact stack that supports quantitative, qualitative, and robustness-oriented analysis rather than a single isolated benchmark number.

1 Introduction

Reading text in natural images remains a central failure mode for otherwise capable vision-language systems. The TextVQA benchmark was introduced precisely to stress this gap: a successful model must not only parse scene text, but also integrate it with visual context and question semantics to produce a short, exact answer [Singh et al., 2019]. In practice, this setting is harder than generic image question answering because many correct responses depend on OCR fidelity, prompt format, answer normalization, and the model's ability to reason over text that is sparse, noisy, or visually embedded in cluttered scenes.

At the same time, the open-source VLM ecosystem has matured enough that a serious empirical study is now feasible without closed commercial models. Architectures such as BLIP-2 [Li et al., 2023], LLaVA [Liu et al., 2023], and Qwen2.5-VL [Qwen Team, 2025] offer strong multimodal priors at parameter scales that can still be evaluated and adapted in an academic setting. Yet the existence of capable models does not, by itself, solve the experimental-design problem. A naïve study can easily become either too shallow, by comparing only a small number of zero-shot settings, or too diffuse, by

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mixing screening, validation, prompt exploration, and fine-tuning without a clear promotion policy. In TextVQA, where OCR choices matter almost as much as backbone choice, such looseness makes the final conclusions difficult to defend.

This paper therefore adopts a deliberately staged protocol rather than a flat benchmark. We first run broad zero-shot screening on a stratified internal development split, then reserve the official validation set for a narrower set of finalists, and finally conduct a winner-conditioned LoRA study on the selected backbone. The key principle is that each stage should answer a different scientific question: screening compares backbones and OCR-sensitive prompting cheaply, finalist reruns verify which combinations are still competitive on the official split, and the training stage isolates the effect of parameter-efficient adaptation while keeping follow-up ablations attached to the actual winning LoRA family rather than to an arbitrary default. The result is an experiment that is broader than a single-tuned-model class project, but still structured enough to remain methodologically coherent.

Our study is also designed around realistic compute constraints rather than idealized cluster access. All canonical evaluation runs are executed locally in a resumable queue on a single Apple-Silicon workstation, while only the final LoRA matrix is moved to rented remote GPUs under a separate run namespace. This separation is not merely operational. It preserves a clean experimental boundary between the local zero-shot/finalist evidence and the remote training evidence, reducing the risk that nominally identical runs mix hardware regimes, caches, or checkpoints in ways that complicate interpretation.

The paper makes the following contributions. First, it introduces a reproducible funnel protocol for benchmarking open-source VLMs on TextVQA without exhausting the official validation split during exploration. Second, it treats OCR as a first-class experimental variable across zero-shot prompting, training-time ablation, and post-hoc robustness analysis. Third, it develops a winner-conditioned LoRA study in which core family selection is based on mean internal-development `eval_loss` across seeds, whereas final tuned-model promotion is based on post-training answer accuracy. This separation is important: it keeps the training-stage search objective distinct from the final paper-facing evaluation objective. Finally, the project delivers a complete implementation pipeline that couples layered configuration, resumable execution, remote multi-GPU orchestration, and progress reporting, making the empirical claims auditable rather than anecdotal.

Section 2 situates the study with respect to TextVQA, open-source VLMs, and parameter-efficient adaptation. Section 3 then formalizes the staged experimental protocol and systems boundary that govern the full study.

2 Related Work

2.1 TextVQA and text-grounded visual question answering

TextVQA was introduced to measure a model's ability to answer questions whose solution depends on reading text embedded in real-world images [Singh et al., 2019]. Unlike generic VQA, the benchmark frequently requires exact recognition of sparse text, reasoning over OCR tokens, and answer generation under strict string-matching evaluation. This makes the problem particularly sensitive to small prompt decisions, OCR formatting, and answer normalization. For an empirical paper, the implication is straightforward: a convincing TextVQA study must treat text handling as part of the research question rather than as an implementation footnote.

That requirement shapes the present study in two ways. First, OCR-aware prompting is treated as a structured experimental variable throughout zero-shot evaluation rather than as a single fixed preprocessing choice. Second, OCR remains part of the fine-tuning analysis through explicit on/off follow-up runs. This design choice is motivated by the benchmark itself: if scene text is central to the task, then conclusions about backbone quality or adaptation quality are incomplete unless OCR usage is interrogated directly.

2.2 Open-source vision-language models

Open-source VLMs have advanced rapidly through increasingly effective combinations of pretrained vision encoders, pretrained language models, and instruction-tuning pipelines. BLIP-2 demonstrated that strong multimodal transfer can be achieved by bootstrapping from frozen unimodal components rather than retraining the full stack from scratch [Li et al., 2023]. LLaVA showed that visual instruction tuning can substantially improve general-purpose multimodal behavior with a comparatively lightweight alignment stage [Liu et al., 2023]. More recent model families, including Qwen2.5-VL [Qwen Team, 2025] and InternVL-style systems, have pushed open-source multimodal performance far beyond early proof-of-concept systems, making it realistic to benchmark several families within one study.

However, strong checkpoints alone do not guarantee a rigorous comparison. Public model reports typically emphasize aggregate benchmark leaderboards, while a project like ours must instead decide how to compare models under a fixed, transparent protocol. In particular, an academic benchmark on TextVQA must choose how much prompt breadth to allow, which split should be used for screening, and when it is appropriate to spend the official validation budget. Our funnel design can be understood as a methodological answer to that problem: broad internal-development screening is used to expose prompt and backbone sensitivity, while the official validation split is deliberately held back for a much smaller set of finalists and final tuned-model checks.

2.3 Parameter-efficient adaptation

The fine-tuning stage in this project is built around Low-Rank Adaptation (LoRA) [Hu et al., 2022], which remains the most practical parameter-efficient adaptation method for a study that aims to compare several adaptation settings rather than optimize a single maximal model. LoRA is especially appropriate here for three reasons. First, it makes a multi-run ablation feasible within a limited compute budget. Second, it allows adaptation choices such as target-module strategy and rank to be varied systematically. Third, it keeps the training question close to the paper's actual scope: we are not trying to establish a new state of the art on TextVQA, but rather to understand how far a strong open-source backbone can be pushed under a controlled and reproducible adaptation budget.

Most importantly, we treat LoRA not as a one-off fine-tuning step but as a structured study. The core matrix compares attention-only and all-linear target strategies, rank 16 versus 32, and seed stability. The follow-up runs then ask higher-level questions that the core matrix alone cannot answer: whether OCR conditioning materially changes adapted performance, and whether additional data continues to improve the selected family. This winner-conditioned design is more informative than a single tuned checkpoint and more defensible than ablations attached to a family that never actually won the initial training comparison.

3 Study Design and Experimental Protocol

3.1 Task and data protocol

The study follows the official TextVQA split structure of approximately 34.6k training examples, 5k validation examples, and 5.73k test examples [Singh et al., 2019]. We reserve the official validation split for finalist reruns, appendix experiments, and final tuned-model checks. To avoid exhausting this split during exploratory screening, we materialize a separate 2,000-example internal development set from the training split. This internal-dev set is stratified rather than sampled naively, with explicit control over question prefix, answer-length bucket, and OCR-token-count bucket. The remaining training examples form the training pool for the LoRA study.

This split policy serves two methodological goals. First, it allows broad model and prompt exploration without implicitly overfitting the official validation set. Second, it makes the final validation comparisons more meaningful, because they are reserved for a shortlist that has already survived a cheaper but structured screening stage. The protocol therefore trades a small amount of up-front split engineering for a much cleaner evaluation narrative.

3.2 Model pool and prompt settings

The canonical zero-shot benchmark pool contains four open-source VLM families and one non-neural OCR lexical baseline. The four reportable VLM families are Qwen2.5-VL-3B, InternVL2.5-4B, LLaVA-Phi-3-mini, and BLIP-2 OPT-2.7B. The OCR lexical baseline is included not as a competitor to modern VLMs, but as a calibration point that helps distinguish genuine multimodal reasoning from question-answer overlap driven by visible text alone.

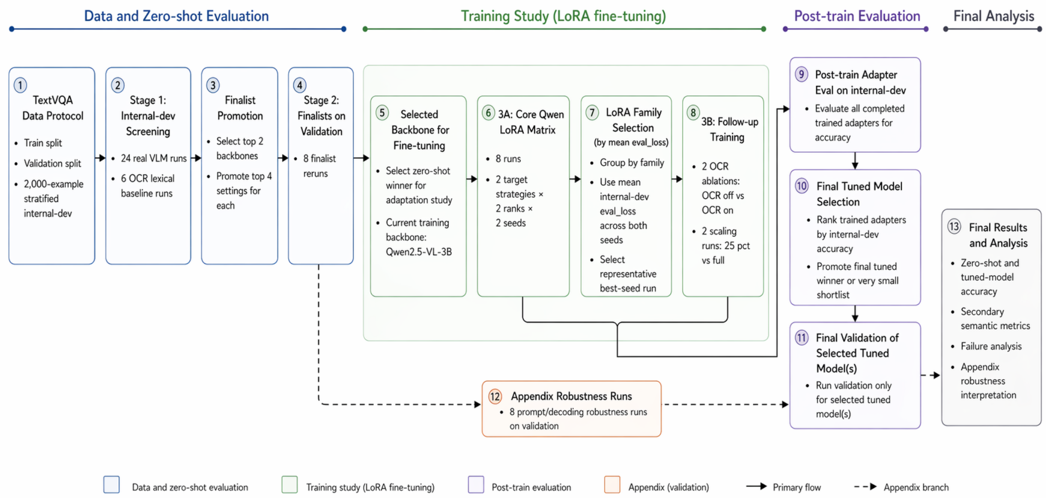

Figure 1: End-to-end experiment pipeline for the current TextVQA study. The protocol follows a staged funnel: stratified internal-dev screening, finalist reruns on the official validation split, backbone selection for LoRA fine-tuning, an 8-run core Qwen LoRA matrix, and 4 winner-conditioned follow-up training runs. Training follow-ups are chosen by mean internal-dev `eval_loss` across core seeds, whereas completed trained adapters are later ranked by internal-dev accuracy before promoting only the final tuned model, or a very small shortlist, to validation. Appendix robustness runs remain a secondary validation branch that supports the final analysis without altering the main winner-selection funnel.

Each real VLM is screened under six prompt settings:

1. `plain`,
2. `short_answer`,
3. `ocr_copy_first`,
4. `ocr_injected`,
5. `ocr_injected_normalized`,
6. `ocr_fused`.

This yields 24 real VLM screening runs (4 backbones $\times$ 6 settings) plus 6 OCR-baseline runs. The design deliberately treats OCR prompting as a first-class variable, because TextVQA performance is often determined as much by how text is exposed to the model as by which backbone is chosen.

3.3 Funnel evaluation protocol

The first evaluation stage performs broad screening on the internal-dev split. Its purpose is not to produce final leaderboard claims, but to estimate which backbone–prompt combinations are competitive enough to deserve the more expensive validation budget. From this stage, we promote the top two backbones and the top four settings for each, producing eight finalist reruns on the official validation split. A separate appendix branch contains eight additional prompt and stress-test evaluations that are intended to strengthen the analysis section rather than alter the main winner-selection logic.

This funnel design is central to the paper's methodology. It prevents the study from becoming either too small to be informative or too loose to be interpretable. In particular, it allows prompt sensitivity to be measured early, but avoids the common mistake of reporting many exploratory validation runs as if they carried the same evidentiary weight as a small finalist set. The backbone selected for fine-tuning is therefore not the winner of a single flat leaderboard, but the survivor of a staged promotion policy.

3.4 Winner-conditioned LoRA training study

The adaptation path in this project is LoRA fine-tuning on Qwen2.5-VL-3B. The training study has twelve runs in total. The core matrix contains eight runs spanning two target-module strategies, two ranks, and two random seeds. Concretely, it compares attention-only versus all-linear target sets and rank 16 versus 32, with seed replication to estimate whether observed differences are stable or fragile.

The remaining four runs are follow-ups conditioned on the completed core matrix rather than fixed in advance to an arbitrary default family. The first pair is an OCR-conditioning ablation that toggles OCR on and off while holding the selected LoRA family fixed. The second pair is a data-scaling comparison between a smaller fixed-budget subset and the full training-remainder pool. Training-stage winner selection is based on mean internal-dev `eval_loss` across both seeds within a family, with the best-seed run used as the representative configuration for writing the winner-conditioned override. Post-training model promotion is handled separately: once trained adapters are evaluated on internal-dev for answer accuracy, only the final tuned winner, or a very small shortlist, is promoted to official validation.

This separation between training-stage family selection and post-training accuracy-based promotion is intentional. The former is a control mechanism for deciding which family should receive follow-up training budget, while the latter is the paper-facing decision rule for selecting the tuned model that deserves final validation evidence. Keeping these roles distinct makes the adaptation study easier to defend and prevents the follow-up phase from being attached to a family that won only by default.

3.5 Metrics, systems boundary, and reproducibility

The headline metric is exact-match TextVQA accuracy. In addition, we report secondary semantic and overlap metrics that capture partial-answer quality beyond exact string matching, including BLEU, METEOR, ROUGE-L, token-level F1, and precision/recall. These metrics do not replace exact-match accuracy, which remains the primary paper-facing score, but they are useful for understanding whether near-miss generations or paraphrastic answers carry signal that exact matching discards. We deliberately keep this metric stack offline and reproducible from saved prediction files, rather than introducing an external API-backed judge whose scores would depend on a separate proprietary service boundary.

The implementation boundary is equally deliberate. All canonical zero-shot, finalist, and appendix evaluations are executed locally on a single Apple-Silicon workstation with resumable queueing and append-only logs. Only the final LoRA matrix is rerun on remote H100 GPUs, and it is rerun from scratch under a distinct run namespace rather than mixed with earlier local pilot training attempts. This preserves a clean hardware and checkpoint boundary between evaluation evidence and training evidence. Reproducibility is further supported through layered TOML configuration, materialized manifests, resumable training checkpoints, automated progress reporting, and remote multi-GPU orchestration with explicit phase boundaries. The result is an experiment that is not only ambitious in scope, but also inspectable in its operational details.

4 Results

4.1 Zero-shot screening and finalist reruns

The internal-dev screening matrix immediately separates backbone quality from prompt sensitivity. Figure 2 shows that Qwen2.5-VL-3B dominates the full screening stage, with its best configuration (`short_answer`) reaching 81.05% accuracy and its weakest OCR-heavy variant still remaining competitive. The ranking is not a trivial prompt artifact: even after allowing six prompt/OCR settings per backbone, Qwen remains clearly ahead of LLaVA-Phi-3-mini, BLIP-2 OPT-2.7B, InternVL2.5-4B, and the OCR lexical baseline. The same figure also shows that prompt sensitivity is model dependent. Qwen prefers concise answer formatting, whereas LLaVA benefits more from OCR-normalized prompting; by contrast, BLIP-2 remains weak across all prompt variants, and InternVL2.5-4B never becomes competitive under the study protocol.

The finalist stage confirms that the screening winner is not an artifact of the internal-dev split. Table 1 shows that the four promoted Qwen settings occupy the top four validation positions, led again by `short_answer` at 78.72% accuracy. The strongest non-Qwen finalist, LLaVA-Phi-3-mini with

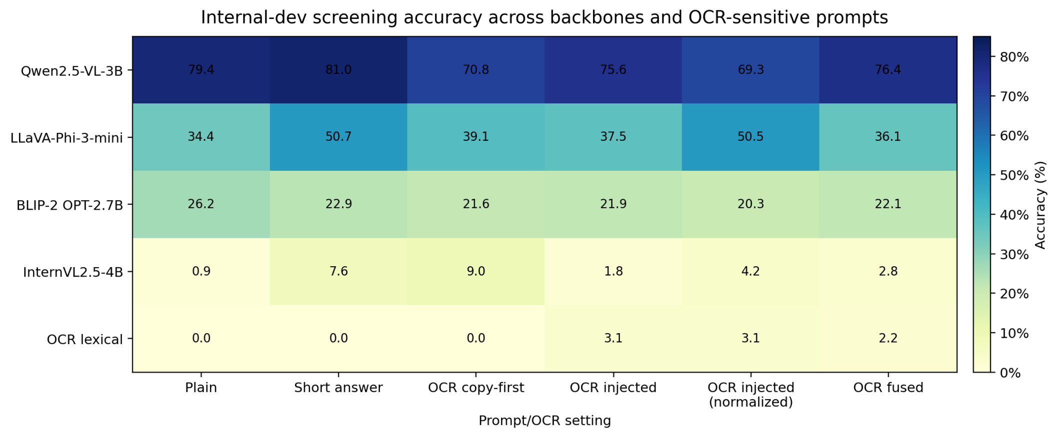

Figure 2: Internal-dev screening accuracy across the full backbone–prompt matrix. Qwen2.5-VL-3B is the only family that remains strong across all OCR-sensitive prompt variants, and its short-answer prompt is the best zero-shot configuration overall. The heatmap also makes the model-dependent nature of OCR prompting visible: OCR normalization helps LLaVA markedly more than it helps Qwen, while the lexical OCR baseline remains near zero throughout.

Stage	Model / setting	Acc.	Cons.	F1	BLEU	METEOR	ROUGE-L
Best screening	Qwen2.5-VL-3B (short answer)	81.05	77.20	70.24	13.48	51.26	72.38
Best screening	LLaVA-Phi-3-mini (short answer)	50.70	46.93	45.74	7.29	36.24	48.03
Best screening	BLIP-2 OPT-2.7B (plain)	26.25	23.50	23.62	2.58	25.13	24.83
Best screening	InternVL2.5-4B (ocr copy first)	9.05	8.78	11.84	0.14	8.26	19.77
Best screening	OCR lexical (ocr injected)	3.15	2.83	10.45	0.59	11.00	11.43
Finalist	Qwen2.5-VL-3B (short answer)	78.72	74.45	70.42	14.88	48.08	73.45
Finalist	Qwen2.5-VL-3B (plain)	77.18	72.91	69.47	14.64	47.81	72.78
Finalist	Qwen2.5-VL-3B (ocr fused)	75.56	70.93	67.37	14.14	46.88	70.84
Finalist	Qwen2.5-VL-3B (ocr injected)	74.42	69.99	66.69	13.92	45.93	70.26
Finalist	LLaVA-Phi-3-mini (ocr injected normalized)	50.36	46.98	48.72	7.26	35.49	52.34
Finalist	LLaVA-Phi-3-mini (short answer)	46.70	43.42	45.08	7.68	33.06	47.99
Finalist	LLaVA-Phi-3-mini (ocr injected)	40.68	37.65	41.66	5.13	32.26	46.80
Finalist	LLaVA-Phi-3-mini (ocr copy first)	36.86	34.42	41.81	7.07	32.97	44.67
Tuned winner	Qwen2.5-VL-3B + LoRA (all-linear r16 seed13)	84.76	80.39	74.92	14.26	50.66	77.56

Table 1: Main benchmark results. The table reports the best internal-dev screening configuration for each backbone, all promoted validation finalists, and the final tuned model. Exact-match accuracy is the primary score; consensus accuracy, F1, BLEU, METEOR, and ROUGE-L provide secondary semantic context. The tuned Qwen adapter becomes the strongest overall validation system.

OCR-normalized prompting, reaches only 50.36%, leaving a 28.36-point gap to the best Qwen finalist. This margin is too large to interpret as prompt noise. Instead, it indicates that under the current evaluation boundary, backbone quality dominates the ranking and that OCR-aware prompting changes ordering within a family more than it changes ordering across families.

4.2 Winner-conditioned LoRA adaptation

The adaptation stage delivers a real validation gain rather than a cosmetic retuning of the zero-shot winner. Figure 3 reports the headline comparison: the final tuned model, `all-linear-r16-seed13`, reaches 84.76% validation accuracy, compared with 78.72% for the strongest zero-shot finalist, for an absolute improvement of 6.04 points. The same comparison also improves consensus accuracy by 5.95 points, F1 by 4.50 points, and ROUGE-L by 4.11 points. The 95% confidence intervals for the two validation accuracies do not overlap, which makes the gain difficult to dismiss as evaluation noise.

The internal-dev adapter sweep in Figure 3 and Table 2 also reveals an important methodological distinction. Training-stage family selection chose `all-linear-r32` as the winner-conditioned follow-up family because it had the lowest mean internal-dev `eval_loss` across seeds. However, the highest post-train internal-dev accuracy among the completed core adapters came from

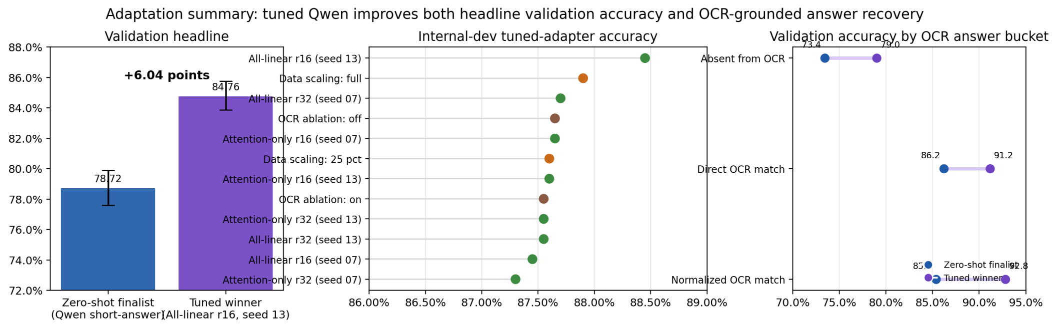

Figure 3: Adaptation summary. Left: the final tuned Qwen model improves validation accuracy by 6.04 points over the strongest zero-shot finalist. Middle: internal-dev post-train accuracy across the full 12-run LoRA study, including the core matrix and winner-conditioned follow-ups. Right: validation gains are largest for OCR-grounded answers, especially those that require normalized OCR matching.

Group	Configuration	Eval loss	Acc.	Cons.	F1	METEOR	ROUGE-L
Core matrix	All-linear r16 (seed 13)	0.5759	88.45	85.03	76.72	55.05	78.34
Data scaling	Data scaling: full	0.5576	87.90	84.80	76.71	54.25	78.22
Core matrix	All-linear r32 (seed 07)	0.5654	87.70	83.92	75.96	54.77	77.44
Core matrix	Attention-only r16 (seed 07)	0.5787	87.65	84.05	76.48	54.92	77.93
OCR ablation	OCR ablation: off	0.5715	87.65	84.13	76.10	55.19	77.64
Core matrix	Attention-only r16 (seed 13)	0.5769	87.60	84.38	76.49	54.61	78.04
Data scaling	Data scaling: 25 pct	0.5724	87.60	84.02	75.98	54.71	77.57
Core matrix	All-linear r32 (seed 13)	0.5833	87.55	84.42	76.22	54.06	77.73
Core matrix	Attention-only r32 (seed 13)	0.5739	87.55	84.27	76.72	54.60	78.28
OCR ablation	OCR ablation: on	0.5672	87.55	84.25	75.77	54.85	77.44
Core matrix	All-linear r16 (seed 07)	0.5803	87.45	84.20	76.23	54.71	77.87
Core matrix	Attention-only r32 (seed 07)	0.5798	87.30	83.65	76.17	54.72	77.70

Table 2: Post-train internal-dev results for the full LoRA study. Core-matrix rows compare target strategy, rank, and seed. Follow-up rows inherit the winner-conditioned family and probe OCR conditioning or data scaling. The table makes the distinction between training-stage `eval_loss` and post-train answer accuracy explicit.

`all-linear-r16-seed13` at 88.45%. This is not a contradiction; it is evidence that training loss is a useful control signal for allocating follow-up budget, but not a perfect substitute for the paper-facing metric of answer accuracy. The separation between family selection and final model promotion is therefore justified empirically, not just procedurally.

Seed stability in the core matrix is also stronger than the raw leaderboard might suggest. Averaging the two seeds within each family, `all-linear-r16` reaches 87.95% internal-dev accuracy, compared with 87.63% for `all-linear-r32`, 87.63% for `attn-r16`, and 87.43% for `attn-r32`. More importantly, the seed ranges are small: 1.00 point for `all-linear-r16`, 0.15 for `all-linear-r32`, 0.05 for `attn-r16`, and 0.25 for `attn-r32`. The study therefore does not hinge on a single anomalous seed.

The follow-up runs sharpen the interpretation of the core matrix rather than merely extending it. The OCR ablation pair differs by only 0.10 internal-dev accuracy points (87.65% without OCR conditioning versus 87.55% with OCR conditioning), suggesting that once the model is adapted, explicit OCR toggling matters less than it did in zero-shot prompting. The data-scaling pair is more informative: moving from the 25 pct setting to the full training-remainder run raises internal-dev accuracy from 87.60% to 87.90%. This gain is modest, but directionally consistent with the training-stage `eval_loss` improvement and supports the choice to spend the final training budget on the full-data configuration.

Branch	Setting	Acc.	Cons.	F1	BLEU	METEOR	ROUGE-L
Prompt study	question routed	75.82	71.41	67.95	13.76	46.39	71.45
Stress test	generation cap 12	74.16	69.53	66.80	15.53	47.25	70.27
Prompt study	ocr copy first	74.00	69.55	67.20	14.55	47.11	70.95
Prompt study	baseline finalist	73.08	68.56	66.37	14.70	46.94	69.92
Stress test	ocr cap 16	73.02	68.53	66.33	14.66	46.89	69.94
Stress test	ocr cap 64	72.84	68.27	66.27	14.72	46.87	69.84
Prompt study	answer format constrained	71.66	67.17	64.72	13.94	45.11	68.46
Stress test	generation cap 4	64.76	60.97	61.76	7.86	43.24	65.87

Table 3: Validation robustness results from the appendix branch. These runs are intentionally secondary to the main funnel and are used to probe prompt-routing and decoding sensitivity without changing the core winner-selection logic.

4.3 Appendix robustness runs

The appendix branch strengthens the analysis without altering the main winner-selection funnel. Table 3 shows that the strongest appendix configuration, `question_routed`, reaches 75.82% validation accuracy. This is competitive, but it remains below the finalist-stage `short_answer` configuration at 78.72%, which is exactly the role that the appendix branch should play: it offers stress tests and alternative prompt formulations that enrich the interpretation, but it does not retroactively rewrite the finalist protocol. Among the stress tests, lowering the generation cap to 4 hurts performance severely, falling to 64.76%, while more moderate caps remain close to the finalist baseline. The appendix therefore supports a nuanced conclusion: Qwen is reasonably robust to modest decoding perturbations, but overly aggressive truncation is materially harmful.

5 Analysis and Failure Modes

5.1 Where tuning helps most

The tuned model's gains are not evenly distributed across the validation set. Figure 3 already shows the broad pattern: adaptation helps most when the answer is recoverable from OCR tokens, especially after normalization. Compared with the strongest zero-shot finalist, the tuned model improves validation accuracy by 4.98 points for direct OCR matches and by 7.46 points for answers that become recoverable only after normalization. The gain is still positive when the answer is absent from OCR entirely (+5.57 points), but it is smaller than the gain on normalized OCR-grounded cases. This is a useful result for interpretation. Fine-tuning is not merely making the model more generically capable; it appears to improve how the model uses text that is already present but noisy, fragmented, or only approximately aligned with the canonical answer string.

The OCR-count buckets tell the same story at a different level of granularity. Validation accuracy on OCR-free cases is unchanged at 66.09%, whereas OCR-rich cases improve substantially: +4.52 points for 1–5 OCR tokens, +5.85 for 6–15, +8.46 for 16–30, and +9.09 for 31+ tokens. In other words, the strongest adaptation gains occur exactly where TextVQA is most text dependent. This makes the final model more convincing as a TextVQA system rather than merely as a stronger short-answer VQA model.

The answer-length analysis also sharpens the interpretation. The tuned model improves one-token answers by 5.85 points, 2–3-token answers by 6.31 points, and 4+-token answers by 7.09 points. Even after tuning, however, long answers remain the hardest category: 76.69% validation accuracy versus 85.63% for one-token answers. The gain is therefore real but not saturating. Parameter-efficient tuning helps on harder answer forms, yet does not eliminate the structural difficulty of longer free-form answers.

5.2 Qualitative evidence

Figure 4 makes the qualitative pattern concrete. In the sparse-OCR watch example, the zero-shot finalist over-predicts the visible time string rather than answering the binary question, while the tuned model emits the short categorical answer required by the task. In the digital-clock example, zero-shot

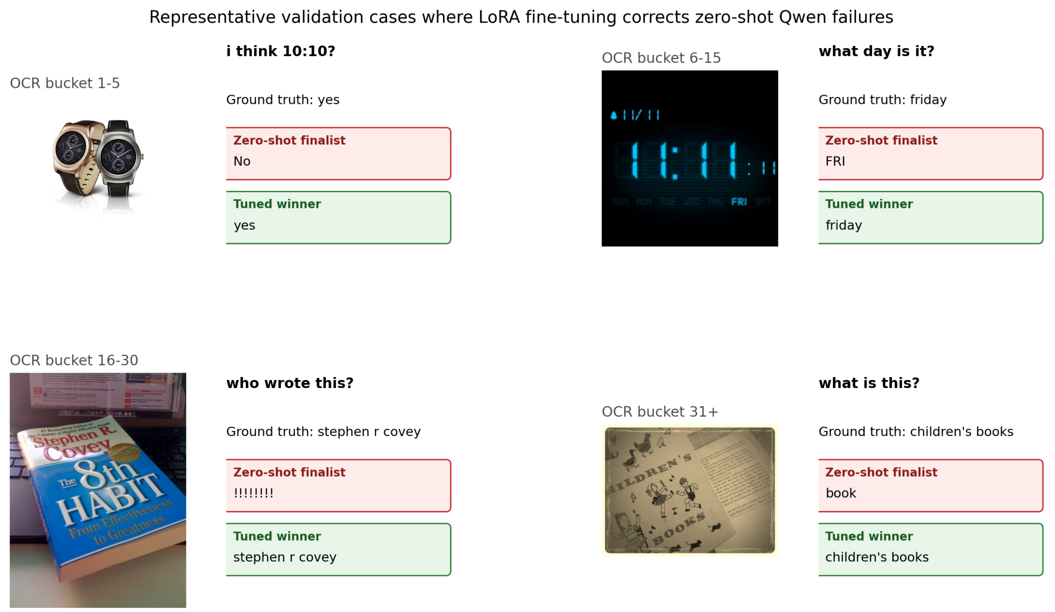

Figure 4: Representative validation cases where LoRA fine-tuning corrects failures made by the zero-shot Qwen finalist. The selected examples span OCR buckets from sparse text to dense text and illustrate three recurring recovery patterns: stricter answer-length control (top left), more faithful OCR grounding (top right), and better normalization or denoising of noisy book-cover text (bottom row).

Qwen partially reads the day token but returns an abbreviated form that fails exact matching, whereas the tuned model normalizes it to the expected answer. The book-cover examples are even more revealing: the tuned model recovers author and title strings from cluttered or degraded text where the zero-shot model either hallucinates punctuation or collapses the answer to an underspecified noun. These cases are consistent with the bucket analysis above. The fine-tuned system is not simply more verbose or more aggressive; it is more disciplined in turning OCR evidence into task-compatible short answers.

5.3 Failure modes and limits

The study also exposes limits that remain after tuning. First, OCR-free questions remain a relative weakness. The tuned model does not improve the zero-shot finalist on the zero-OCR bucket, which suggests that the observed gains are genuinely tied to text-grounded reasoning rather than to a uniform shift in visual question answering quality. Second, despite strong overall improvements, the tuned system is still materially worse on longer answers and on some semantically ambiguous prompts. Third, the OCR ablation follow-up indicates that explicit OCR injection is not the sole driver of post-train gains; once the LoRA adapter is learned, the model appears to internalize part of the text-grounding benefit, but this also makes it harder to isolate exactly which adaptation parameters are compensating for OCR-format differences.

There is also an important methodological limit. The final paper reports automatic semantic metrics such as BLEU, METEOR, and ROUGE-L alongside token-overlap measures, but it does not include an external API-backed LLM-as-a-Judge score. This omission is deliberate rather than accidental: the current artifact stack is designed to be reproducible from saved predictions without introducing a second proprietary evaluation boundary. That choice makes the paper's evidence easier to audit, but it also means that semantically tolerant judgment remains a natural extension for future work.

Finally, the adaptation story is intentionally narrow. We tune only the selected Qwen backbone and only with LoRA. That decision is scientifically coherent with the funnel design and with the available compute budget, but it leaves open at least two follow-up questions: whether the same winner-conditioned protocol would select a different adaptation family on a larger backbone, and whether a richer parameter-efficient method would preserve the same OCR-grounded gains without increasing instability or cost.

6 Conclusion

This paper presented a staged TextVQA study that was designed to be broader than a single tuned-model report while remaining methodologically disciplined under a realistic compute budget. The zero-shot funnel established Qwen2.5-VL-3B as the strongest open-source backbone in the benchmark pool, the finalist reruns showed that this conclusion survived transfer to the official validation split, and the winner-conditioned LoRA study converted that backbone advantage into a tuned validation score of 84.76%. The resulting gain over the strongest zero-shot finalist is large enough to matter substantively, but the more important contribution is arguably structural: the study shows how to combine broad screening, validation restraint, parameter-efficient adaptation, follow-up ablations, and reproducible orchestration inside one coherent protocol.

The empirical findings are also more specific than a single leaderboard improvement. Prompting matters, but mostly within a backbone rather than across backbones. Fine-tuning helps, but it helps most on OCR-grounded questions rather than on OCR-free ones. Training-stage loss is a useful selection signal for follow-up allocation, but not a perfect proxy for final answer accuracy. OCR toggling matters less after adaptation than it does in zero-shot prompting, while additional training data continues to help, albeit modestly, even after the main family decision has already been made. Together, these observations make the final tuned model more interpretable than a generic "best checkpoint" claim.

From a project-design perspective, the work also illustrates a practical template for academic multimodal studies. The key ingredients are not exotic: a stratified internal-dev split, a clear promotion rule, a hard boundary between local evaluation and remote training, a winner-conditioned follow-up phase, and artifact-level reproducibility. Yet in combination, these decisions make the final evidence substantially easier to defend. Future extensions should add a semantically richer learned judge, expand adaptation beyond a single backbone, and test whether the same funnel protocol transfers to adjacent OCR-heavy benchmarks. Even without those extensions, the current study already supports a clear conclusion: carefully adapted open-source VLMs can become materially stronger TextVQA systems, provided that the experimental protocol is rigorous enough to distinguish real gains from prompt noise and selection leakage.

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